

DOCKER COMPOSE PRODUCTION SETUP

For: 400k Users in Cameroon | Concurrent: 1,000 | Date: 8/17/2026 | Status: PRODUCTION-READY

DOCKER COMPOSE PRODUCTION SETUP

docker-compose.yml (Complete)

version: "3.8"

services:

nginx:

image: nginx:alpine

container_name: videocall_nginx

ports:

- "80:80"

- "443:443"

volumes:

- ./nginx.conf:/etc/nginx/nginx.conf:ro

- ./certs:/etc/nginx/certs:ro

- ./admin-ui/build:/usr/share/nginx/admin:ro

depends_on:

- node-api

- mediasoup-1

networks:

- videocall

restart: unless-stopped

node-api:

build: ./backend/api

container_name: videocall_api

environment:

- NODE_ENV=production

- PORT=3000

- DATABASE_URL=postgresql://videocall:\${POSTGRES_PASSWORD}@postgres:5432/videocall

- REDIS_URL=redis://redis:6379

- JWT_SECRET=\${JWT_SECRET}

- MEDIASOUP_INSTANCES=mediasoup-1,mediasoup-2

volumes:

- /data/videocall/recordings:/app/recordings

depends_on:

- postgres

- redis

networks:

- videocall

restart: unless-stopped

mediasoup-1:

build: ./backend/mediasoup

container_name: videocall_sfu_1

environment:

- INSTANCE_ID=sfu-1
- LISTENING_IP=0.0.0.0
- ANNOUNCING_IP=\${SERVER_IP}
- MIN_PORT=40000
- MAX_PORT=40100
- REDIS_URL=redis://redis:6379

ports:

- "40000-40100:40000-40100/udp"

volumes:

- /dev/shm:/dev/shm

networks:

- videocall

restart: unless-stopped

mediasoup-2:

build: ./backend/mediasoup

container_name: videocall_sfu_2

environment:

- INSTANCE_ID=sfu-2
- LISTENING_IP=0.0.0.0
- ANNOUNCING_IP=\${SERVER_IP}
- MIN_PORT=40100
- MAX_PORT=40200
- REDIS_URL=redis://redis:6379

ports:

- "40100-40200:40100-40200/udp"

volumes:

- /dev/shm:/dev/shm

networks:

- videocall

restart: unless-stopped

recording:

build: ./backend/recording

container_name: videocall_recording

environment:

- DATABASE_URL=postgresql://videocall:\${POSTGRES_PASSWORD}@postgres:5432/videocall
- REDIS_URL=redis://redis:6379

volumes:

- /data/videocall/recordings:/app/recordings

depends_on:

- postgres
- redis

networks:

- videocall

restart: unless-stopped

postgres:

image: postgres:14-alpine

container_name: videocall_postgres

environment:

- POSTGRES_DB=videocall
- POSTGRES_USER=videocall
- POSTGRES_PASSWORD=\${POSTGRES_PASSWORD}

volumes:

- /data/videocall/postgres-data:/var/lib/postgresql/data
- ./schema.sql:/docker-entrypoint-initdb.d/schema.sql

networks:

- videocall

restart: unless-stopped

healthcheck:

test: ["CMD-SHELL", "pg_isready -U videocall"]

interval: 10s

timeout: 5s

retries: 5

redis:

image: redis:7-alpine

container_name: videocall_redis

volumes:

- /data/videocall/redis-data:/data

networks:

- videocall

restart: unless-stopped

healthcheck:

test: ["CMD", "redis-cli", "ping"]

interval: 10s

timeout: 5s

retries: 5

networks:

videocall:
driver: bridge

`.env.example`

Server Configuration

SERVER_IP=<your-cameroon-server-public-ip>
DOMAIN=videocall.example.com

Database

POSTGRES_PASSWORD=<generate-strong-password>
DATABASE_URL=postgresql://videocall:\${POSTGRES_PASSWORD}@postgres:5432/videocall

Redis

REDIS_URL=redis://redis:6379

JWT

JWT_SECRET=<generate-64-char-random-string>
JWT_EXPIRES_IN=900

Mediasoup

MEDIASOUP_INSTANCES=mediasoup-1,mediasoup-2
MEDIASOUP_RTC_PORT_MIN=40000
MEDIASOUP_RTC_PORT_MAX=40200

TURN Server (Self-hosted Coturn)

TURN_SERVER_IP=<cameroon-coturn-server-ip>
TURN_SERVER_PORT=3478
TURN_USERNAME=videocall
TURN_PASSWORD=<generate-password>

Recording

RECORDINGS_PATH=/data/videocall/recordings
RECORDINGS_RETENTION_DAYS=30
RECORDINGS_BITRATE_KBPS=3000

Admin

ADMIN_EMAIL=admin@videocall.local
ADMIN_PASSWORD=<initial-password>
